

Examen Big Data

SDIA: le 13 Janvier 2025

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Exercice 1 : *HDFS*

- Installez **HDFS** en utilisant **Docker**. Fournissez les commandes Docker utilisées.

```
# Créer un réseau Docker pour notre cluster HDFS
sudo docker network create --driver bridge hadoopnet

# Démarrer le NameNode avec des variables d'environnement appropriées
sudo docker run -d --name namenode --network hadoopnet \
  -p 9870:9870 -p 9000:9000 \
  -e CLUSTER_NAME=hadoop-cluster \
  -e CORE_CONF_fs_defaultFS=hdfs://namenode:9000 \
  -e CORE_CONF_hadoop_http_staticuser_user=root \
  -e HDFS_CONF_dfs_webhdfs_enabled=true \
  -e HDFS_CONF_dfs_permissions_enabled=false \
  bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8

# Démarrer les DataNodes avec la configuration appropriée
sudo docker run -d --name datanode1 --network hadoopnet \
  -e SERVICE_PRECONDITION="namenode:9870" \
  -e CORE_CONF_fs_defaultFS=hdfs://namenode:9000 \
  -e CORE_CONF_hadoop_http_staticuser_user=root \
  bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8

sudo docker run -d --name datanode2 --network hadoopnet \
  -e SERVICE_PRECONDITION="namenode:9870" \
  -e CORE_CONF_fs_defaultFS=hdfs://namenode:9000 \
  -e CORE_CONF_hadoop_http_staticuser_user=root \
  bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8
```

- Configurez un cluster **HDFS** simple avec un **NameNode** et deux **DataNodes**.

```
# Vérifier que les conteneurs sont en cours d'exécution
sudo docker ps

# Vérifier l'état du cluster HDFS
sudo docker exec -it namenode hdfs dfsadmin -report
```

- Importez un fichier texte (CSV) contenant des informations de ventes (colonnes : transaction_id, user_id, product_id, timestamp, amount) dans **HDFS**. Montrez les commandes utilisées pour importer le fichier et vérifier son existence dans HDFS.

```
# Création du fichier de ventes
cat << 'EOF' > sales.csv
transaction_id,user_id,product_id,timestamp,amount
1001,user123,prod456,2024-03-20 10:30:00,150.50
1002,user124,prod789,2024-03-20 10:35:00,75.25
1003,user125,prod456,2024-03-20 10:40:00,200.00
1004,user126,prod123,2024-03-20 10:45:00,99.99
1005,user127,prod456,2024-03-20 10:50:00,175.75
EOF

# Copier le fichier dans le conteneur NameNode
sudo docker cp sales.csv namenode:/tmp/

# Créer le répertoire et importer le fichier dans HDFS
sudo docker exec -it namenode bash
hdfs dfs -mkdir -p /data/sales
hdfs dfs -put /tmp/sales.csv /data/sales/
```

- Listez le contenu du répertoire **HDFS**, lisez le début du fichier importé, puis copiez ce fichier vers un autre répertoire dans **HDFS**. Fournissez les commandes et leurs sorties.

```
# Lister le contenu du répertoire
hdfs dfs -ls /data/sales/

# Lire les premières lignes du fichier
hdfs dfs -cat /data/sales/sales.csv | head -n 3
```

- Configurez la réplication du fichier importé avec un **facteur de réplication de 3**. Vérifiez et montrez que la réplication est correctement configurée.

```
# Configurer le facteur de réplication
hdfs dfs -setrep 3 /data/sales/sales.csv

# Vérifier la configuration de réplication
hdfs dfs -ls -R /data/sales/
```

- Image HDFS :

```
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$ sudo docker network create hadoopnet
[sudo] password for tr:
3239b0f306ac438103d63a0b4ca3051036ed8deb047dcb5875dd25664dcc8946
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$ sudo docker network create hadoopnet
Error response from daemon: network with name hadoopnet already exists
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$
sudo docker run -d --name namenode --network hadoopnet \
    -p 9870:9870 -p 9000:9000 \
    -e CLUSTER_NAME=hadoop-cluster \
    bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8
Unable to find image 'bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8' locally
2.0.0-hadoop3.2.1-java8: Pulling from bde2020/hadoop-namenode
3192219afd04: Pull complete
7127a1d8cccd: Pull complete
883a89599900: Pull complete
77920a3e82af: Pull complete
92329e81aec4: Pull complete
f373218fec59: Pull complete
aa53513fe997: Pull complete
8b1800105b98: Pull complete
c3a84a3e49c8: Pull complete
a65640a64a76: Pull complete
facffb3a6de3: Pull complete
c71a6df73788: Pull complete
73b8c0ccb707: Pull complete
Digest: sha256:51ad9293ec52083c5003ef0aaab00c3dd7d6335ddf495cc1257f97a272cab4c0
Status: Downloaded newer image for bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8
c4fec0bde4b1e89d311e0313334532a8fb764d53378576aa6c7285a3848c8339
```

```
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$ sudo docker run -d --name namenode --network hadoopnet \
    -p 9870:9870 -p 9000:9000 \
    -e CLUSTER_NAME=hadoop-cluster \
    -e CORE_CONF_fs_defaultFS=hdfs://namenode:9000 \
    -e CORE_CONF_hadoop_http_staticuser_user=root \
    -e HDFS_CONF_dfs_webhdfs_enabled=true \
    -e HDFS_CONF_dfs_permissions_enabled=false \
    bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8
d0a4f4c9e35a253e4cf44feac18adcd54e1e9f1c7bcce3247ff57d7cb326b820
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$ sudo docker run -d --name datanode1 --network hadoopnet \
    -e SERVICE_PRECONDITION="namenode:9870" \
    -e CORE_CONF_fs_defaultFS=hdfs://namenode:9000 \
    -e CORE_CONF_hadoop_http_staticuser_user=root \
    bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8
c3da872c8cd89c05f8e3240ce6bd2386f190d95351f09ed0c893dbd6b6780af6
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$ sudo docker run -d --name datanode2 --network hadoopnet \
    -e SERVICE_PRECONDITION="namenode:9870" \
    -e CORE_CONF_fs_defaultFS=hdfs://namenode:9000 \
    -e CORE_CONF_hadoop_http_staticuser_user=root \
    bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8
f2fe676f2f79bffa756e0376763d4e4e8ae7e7c859f07cc62474b18b972c1a5
```

```
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$ sudo docker ps
CONTAINER ID   IMAGE                                     NAMES
f2fe676f2f79   bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8   datanode2
c3da872c8cd8   bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8   datanode1
d0a4f4c9e35a   bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8   namenode
0/tcp, :::9870->9870/tcp   bitnaml/kafkalatest   kafka
528eeb2373d9   bitnaml/zookeeper:latest   zookeeper
2cb09c68a1a1   bitnaml/zookeeper:latest   zookeeper
p, 8080/tcp

tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$ sudo docker exec -it namenode hdfs dfsadmin -report
Configured Capacity: 24979311808 (232.64 GB)
Present Capacity: 24824334336 (22.37 GB)
DFS Remaining: 24824285184 (22.37 GB)
DFS Used: 49152 (48 KB)
DFS Used%: 0.00%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
Erasure Coded Block Groups:
  Low redundancy block groups: 0
  Block groups with corrupt internal blocks: 0
  Missing block groups: 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0

-----
Live datanodes (2):
```

```

-----
Live datanodes (2):

Name: 172.20.0.3:9866 (datanode1.hadoopnet)
Hostname: c3da872c8cd8
Decommission Status : Normal
Configured Capacity: 124896559104 (116.32 GB)
DFS Used: 24576 (24 KB)
Non DFS Used: 106492747776 (99.18 GB)
DFS Remaining: 12012142592 (11.19 GB)
DFS Used%: 0.00%
DFS Remaining%: 9.62%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 1
Last contact: Mon Jan 13 08:32:19 UTC 2025
Last Block Report: Mon Jan 13 08:31:31 UTC 2025
Num of Blocks: 0

Name: 172.20.0.4:9866 (datanode2.hadoopnet)
Hostname: f2fe676f2f79
Decommission Status : Normal
Configured Capacity: 124896559104 (116.32 GB)
DFS Used: 24576 (24 KB)
Non DFS Used: 106492747776 (99.18 GB)
DFS Remaining: 12012142592 (11.19 GB)
DFS Used%: 0.00%
DFS Remaining%: 9.62%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 1
Last contact: Mon Jan 13 08:32:18 UTC 2025
Last Block Report: Mon Jan 13 08:31:43 UTC 2025
Num of Blocks: 0

```

```

tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exan$ cat << 'EOF' > sales.csv
transaction_id,user_id,product_id,timestamp,amount
1001,user123,prod456,2024-03-20 10:30:00,150.50
1002,user124,prod789,2024-03-20 10:35:00,75.25
1003,user125,prod456,2024-03-20 10:40:00,200.00
1004,user126,prod123,2024-03-20 10:45:00,99.99
1005,user127,prod456,2024-03-20 10:50:00,175.75
EOF
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exan$ sudo docker cp sales.csv namenode:/tmp/
Successfully copied 2.05kB to namenode:/tmp/

tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exan$ sudo docker exec -it namenode bash
root@d0a4f4c9e35a:/#
root@d0a4f4c9e35a:/# hdfs dfs -mkdir -p /data/sales
root@d0a4f4c9e35a:/# hdfs dfs -put /tmp/sales.csv /data/sales/
2025-01-13 08:34:51,110 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
root@d0a4f4c9e35a:/# hdfs dfs -ls /data/sales/
Found 1 items
-rw-r--r-- 3 root supergroup 289 2025-01-13 08:34 /data/sales/sales.csv
root@d0a4f4c9e35a:/# hdfs dfs -cat /data/sales/sales.csv | head -n 3
2025-01-13 08:34:56,355 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
transaction_id,user_id,product_id,timestamp,amount
1001,user123,prod456,2024-03-20 10:30:00,150.50
1002,user124,prod789,2024-03-20 10:35:00,75.25
root@d0a4f4c9e35a:/# hdfs dfs -setrep 3 /data/sales/sales.csv
Replication 3 set: /data/sales/sales.csv
root@d0a4f4c9e35a:/# hdfs dfs -ls -R /data/sales/
-rw-r--r-- 3 root supergroup 289 2025-01-13 08:34 /data/sales/sales.csv
root@d0a4f4c9e35a:/#

```

Exercise 2 : Spring Batch

1. Configuration du projet Spring Batch :

```

<!-- pom.xml -->
<dependencies>
  <dependency>
    <groupId>org.springframework.boot</groupId>

```

```

        <artifactId>spring-boot-starter-batch</artifactId>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-data-jpa</artifactId>
    </dependency>
    <dependency>
        <groupId>com.h2database</groupId>
        <artifactId>h2</artifactId>
        <scope>runtime</scope>
    </dependency>
    <dependency>
        <groupId>org.projectlombok</groupId>
        <artifactId>lombok</artifactId>
    </dependency>
</dependencies>

```

2. Configuration de l'application :

```

# application.properties
spring.application.name=exercice-2
server.port=8082

# H2 Database Configuration
spring.datasource.url=jdbc:h2:mem:salesdb
spring.datasource.driverClassName=org.h2.Driver
spring.datasource.username=sa
spring.datasource.password=
spring.h2.console.enabled=true

# JPA Configuration
spring.jpa.database-platform=org.hibernate.dialect.H2Dialect
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true

# Batch Configuration
spring.batch.jdbc.initialize-schema=always
spring.batch.job.enabled=false

```

3. Modèle de données :

```

@Entity
@Data @NoArgsConstructor @AllArgsConstructor
public class Sale {
    @Id
    private Long transactionId;
    private String userId;
    private String productId;
    @DateTimeFormat(pattern = "yyyy-MM-dd HH:mm:ss")
    private LocalDateTime timestamp;
}

```

```

    private Double amount;
    private String category;
    private Integer quantity;
    private Double unitPrice;
}

@Entity
@Data @NoArgsConstructor @AllArgsConstructor
public class SalesSummary {
    @Id @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;
    private String category;
    private Double totalRevenue;
    private Integer totalProducts;
    private Double averageRevenuePerProduct;
}

```

4. Configuration du Batch :

```

@Configuration
@EnableScheduling
public class BatchConfig {
    @Bean
    public FlatFileItemReader<Sale> reader() {
        FlatFileItemReader<Sale> reader = new FlatFileItemReader<>();
        reader.setResource(new ClassPathResource("sales.csv"));
        reader.setLinesToSkip(1);
        reader.setStrict(false);

        DefaultLineMapper<Sale> lineMapper = new DefaultLineMapper<>();
        DelimitedLineTokenizer tokenizer = new DelimitedLineTokenizer();
        tokenizer.setNames("transactionId", "userId", "productId", "timestamp",
            "amount", "category", "quantity", "unitPrice");

        BeanWrapperFieldSetMapper<Sale> fieldSetMapper = new
        BeanWrapperFieldSetMapper<>();
        fieldSetMapper.setTargetType(Sale.class);

        DefaultConversionService conversionService = new DefaultConversionService();
        conversionService.addConverter(String.class, LocalDateTime.class, source ->
            LocalDateTime.parse(source, DateTimeFormatter.ofPattern("yyyy-MM-dd
            HH:mm:ss")));
        fieldSetMapper.setConversionService(conversionService);

        lineMapper.setLineTokenizer(tokenizer);
        lineMapper.setFieldSetMapper(fieldSetMapper);
        reader.setLineMapper(lineMapper);

        return reader;
    }
}

```

```

@Bean
public SalesProcessor processor() {
    return new SalesProcessor();
}

@Bean
public JpaItemWriter<SalesSummary> writer(EntityManagerFactory
entityManagerFactory) {
    JpaItemWriter<SalesSummary> writer = new JpaItemWriter<>();
    writer.setEntityManagerFactory(entityManagerFactory);
    return writer;
}

@Bean
public Job salesJob(JobRepository jobRepository, Step salesStep) {
    return new JobBuilder("salesJob", jobRepository)
        .start(salesStep)
        .build();
}

@Bean
public Step salesStep(JobRepository jobRepository,
PlatformTransactionManager transactionManager,
FlatFileItemReader<Sale> reader,
SalesProcessor processor,
JpaItemWriter<SalesSummary> writer) {
    return new StepBuilder("salesStep", jobRepository)
        .<Sale, SalesSummary>chunk(10, transactionManager)
        .reader(reader)
        .processor(processor)
        .writer(writer)
        .build();
}
}

```

5. Processeur pour le traitement des données :

```

@Component
public class SalesProcessor implements ItemProcessor<Sale, SalesSummary> {
    private Map<String, SalesSummary> summaryMap = new HashMap<>();

    @Override
    public SalesSummary process(Sale sale) throws Exception {
        if (sale.getQuantity() <= 0 || sale.getUnitPrice() <= 0) {
            return null;
        }

        String category = sale.getCategory();
        double revenue = sale.getQuantity() * sale.getUnitPrice();

        SalesSummary summary = summaryMap.computeIfAbsent(category, k -> new

```

```

SalesSummary());
    summary.setCategory(category);
    summary.setTotalRevenue(summary.getTotalRevenue() == null ? revenue :
summary.getTotalRevenue() + revenue);
    summary.setTotalProducts(summary.getTotalProducts() == null ?
sale.getQuantity() : summary.getTotalProducts() + sale.getQuantity());
    summary.setAverageRevenuePerProduct(summary.getTotalRevenue() /
summary.getTotalProducts());

    return summary;
}
}

```

6. Planification du job :

```

@Component
public class JobScheduler {
    @Autowired
    private JobLauncher jobLauncher;

    @Autowired
    private Job job;

    @Scheduled(cron = "0 0 0 * * ?")
    public void runJob() throws Exception {
        JobParameters parameters = new JobParametersBuilder()
            .addLong("time", System.currentTimeMillis())
            .toJobParameters();
        jobLauncher.run(job, parameters);
    }
}

```

7. Contrôleur REST pour le déclenchement manuel :

```

@RestController
@RequestMapping("/api")
public class JobController {
    @Autowired
    private JobLauncher jobLauncher;

    @Autowired
    private Job job;

    @Autowired
    private SalesSummaryRepository summaryRepository;

    @PostMapping("/job/start")
    public String startJob() throws Exception {
        JobParameters parameters = new JobParametersBuilder()
            .addLong("time", System.currentTimeMillis())

```



```

        .toJobParameters();
        jobLauncher.run(job, parameters);
        return "Job started...";
    }

    @GetMapping("/summaries")
    public List<SalesSummary> getSummaries() {
        return summaryRepository.findAll();
    }
}

```

8. Fichier de données d'exemple :

```

transaction_id,user_id,product_id,timestamp,amount,category,quantity,unit_price
1001,user123,prod456,2024-03-20 10:30:00,150.50,Electronics,2,75.25
1002,user124,prod789,2024-03-20 10:35:00,75.25,Books,1,75.25
1003,user125,prod456,2024-03-20 10:40:00,200.00,Electronics,4,50.00
1004,user126,prod123,2024-03-20 10:45:00,99.99,Books,1,99.99
1005,user127,prod456,2024-03-20 10:50:00,175.75,Electronics,3,58.58

```

9. Résultats de l'exécution :

```

[
  {
    "id": 1,
    "category": "Electronics",
    "totalRevenue": 526.24,
    "totalProducts": 9,
    "averageRevenuePerProduct": 58.47
  },
  {
    "id": 2,
    "category": "Books",
    "totalRevenue": 175.24,
    "totalProducts": 2,
    "averageRevenuePerProduct": 87.62
  }
]

```

- Image Spring Batch :

The Spring Initializr interface shows the following configuration:

- Project:** Maven
- Language:** Java
- Spring Boot:** 3.4.1
- Project Metadata:**
 - Group: ma.enset
 - Artifact: exercice-2
 - Name: exercice-2
 - Description: Exercice 2 par Omar Trizi
 - Package name: ma.enset.exercice2
 - Packaging: Jar
 - Java: 17
- Dependencies:**
 - H2 Database (SQL)
 - Lombok (DEVELOPER TOOLS)
 - Spring Web (WEB)
 - Spring Data JPA (SQL)
 - Spring Batch (IG)

Buttons at the bottom: GENERATE (CTRL + G), EXPLORE (CTRL + SPACE), and a menu icon.

```

11  @Component
12  public class JobScheduler {
13
14      @Autowired
15      private JobLauncher jobLauncher;
16
17      @Autowired
18      private Job salesJob;
19
20      @Scheduled(cron = "0 0 0 * * ?") // Run at midnight every day
21      public void runJob() throws Exception {
22          JobParameters params = new JobParametersBuilder()
23              .addLong("time", System.currentTimeMillis())
24              .toJobParameters();
25
26          jobLauncher.run(salesJob, params);
27      }
28  }

```

```

3/big_data
• tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$ curl http://localhost:8082/api/summaries
[{"id":1,"category":"Electronics","totalRevenue":526.24,"totalProducts":9,"averageRevenuePerProduct":58.471111111111114}, {"id":2,"category":"Books","totalRevenue":175.24,"totalProducts":2,"averageRevenuePerProduct":87.62}, {"id":3,"category":"Electronics","totalRevenue":526.24,"totalProducts":9,"averageRevenuePerProduct":58.471111111111114}, {"id":4,"category":"Books","totalRevenue":175.24,"totalProducts":2,"averageRevenuePerProduct":87.62}, {"id":5,"category":"Electronics","totalRevenue":526.24,"totalProducts":9,"averageRevenuePerProduct":58.471111111111114}]tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/

```

```

• tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$ curl http://localhost:8082/api/summaries
[{"id":1,"category":"Electronics","totalRevenue":526.24,"totalProducts":9,"averageRevenuePerProduct":58.471111111111114}, {"id":2,"category":"Books","totalRevenue":175.24,"totalProducts":2,"averageRevenuePerProduct":87.62}, {"id":3,"category":"Electronics","totalRevenue":526.24,"totalProducts":9,"averageRevenuePerProduct":58.471111111111114}, {"id":4,"category":"Books","totalRevenue":175.24,"totalProducts":2,"averageRevenuePerProduct":87.62}, {"id":5,"category":"Electronics","totalRevenue":526.24,"totalProducts":9,"averageRevenuePerProduct":58.471111111111114}]tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/
• tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$

```

Terminal status bar: main*+ 0 0 0 0 0 Java: Ready

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

HHH10001005: Database info:
    Database JDBC URL [Connecting through datasource 'HikariDataSource (HikariPool-1)']
    Database driver: undefined/unknown
    Database version: 2.3.232
    Autocommit mode: undefined/unknown
    Isolation level: undefined/unknown
    Minimum pool size: undefined/unknown
    Maximum pool size: undefined/unknown
2025-01-13T10:00:44.493+01:00  INFO 216725 --- [exercice-2] [main] o.h.e.t.j.p.i.JtaPlatformInitiator :
HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
2025-01-13T10:00:44.556+01:00  DEBUG 216725 --- [exercice-2] [main] org.hibernate.SQL :
create table sale (transaction_id bigint not null, amount float(53), category varchar(255), product_id varchar(255), qu
ntity integer, timestamp timestamp(6), unit_price float(53), user_id varchar(255), primary key (transaction_id))
Hibernate: create table sale (transaction_id bigint not null, amount float(53), category varchar(255), product_id varcha
r(255), quantity integer, timestamp timestamp(6), unit_price float(53), user_id varchar(255), primary key (transaction_i
d))
2025-01-13T10:00:44.558+01:00  DEBUG 216725 --- [exercice-2] [main] org.hibernate.SQL :
create table sales_summary (id bigint generated by default as identity, average_revenue_per_product float(53), category
varchar(255), total_products integer, total_revenue float(53), primary key (id))
Hibernate: create table sales_summary (id bigint generated by default as identity, average_revenue_per_product float(53)
, category varchar(255), total_products integer, total_revenue float(53), primary key (id))
2025-01-13T10:00:44.564+01:00  INFO 216725 --- [exercice-2] [main] j.LocalContainerEntityManagerFactoryBean :
Initialized JPA EntityManagerFactory for persistence unit 'default'
2025-01-13T10:00:45.046+01:00  WARN 216725 --- [exercice-2] [main] JpaBaseConfiguration$JpaWebConfiguration :
spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed during view rendering. Expl
icly configure spring.jpa.open-in-view to disable this warning
2025-01-13T10:00:45.453+01:00  INFO 216725 --- [exercice-2] [main] o.s.b.a.h2.H2ConsoleAutoConfiguration :
H2 console available at '/h2-console'. Database available at 'jdbc:h2:mem:salesdb'
2025-01-13T10:00:45.533+01:00  DEBUG 216725 --- [exercice-2] [main] .s.JobRegistrySmartInitializingSingleton :
Registering job: salesJob
X  main*+  @ 0 0 0 0  Java: Ready
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Saving step execution before commit: StepExecution: id=1, version=1, name=salesStep, status=STARTED, exitStatus=EXECUTI
NG, readCount=5, filterCount=0, writeCount=5 readSkipCount=0, writeSkipCount=0, processSkipCount=0, commitCount=1, rollb
ackCount=0, exitDescription=
2025-01-13T10:01:11.946+01:00  DEBUG 216725 --- [exercice-2] [nio-8082-exec-1] o.s.batch.repeat.support.RepeatTemplate :
Repeat is complete according to policy and result value.
2025-01-13T10:01:11.946+01:00  DEBUG 216725 --- [exercice-2] [nio-8082-exec-1] o.s.batch.core.step.AbstractStep :
Step execution success: id=1
2025-01-13T10:01:11.951+01:00  INFO 216725 --- [exercice-2] [nio-8082-exec-1] o.s.batch.core.step.AbstractStep :
Step: [salesStep] executed in 176ms
2025-01-13T10:01:11.957+01:00  DEBUG 216725 --- [exercice-2] [nio-8082-exec-1] o.s.batch.core.step.AbstractStep :
Step execution complete: StepExecution: id=1, version=3, name=salesStep, status=COMPLETED, exitStatus=COMPLETED, readCo
unt=5, filterCount=0, writeCount=5 readSkipCount=0, writeSkipCount=0, processSkipCount=0, commitCount=1, rollbackCount=0
2025-01-13T10:01:11.960+01:00  DEBUG 216725 --- [exercice-2] [nio-8082-exec-1] o.s.batch.core.job.AbstractJob :
Upgrading JobExecution status: StepExecution: id=1, version=3, name=salesStep, status=COMPLETED, exitStatus=COMPLETED,
readCount=5, filterCount=0, writeCount=5 readSkipCount=0, writeSkipCount=0, processSkipCount=0, commitCount=1, rollbackC
ount=0, exitDescription=
2025-01-13T10:01:11.960+01:00  DEBUG 216725 --- [exercice-2] [nio-8082-exec-1] o.s.batch.core.job.AbstractJob :
Job execution complete: JobExecution: id=1, version=1, startTime=2025-01-13T10:01:11.740989294, endTime=null, lastUpdat
ed=2025-01-13T10:01:11.741330138, status=COMPLETED, exitStatus=exitCode=COMPLETED;exitDescription=, job=[JobInstance: id
=1, version=0, Job=[salesJob]], jobParameters=[{'time':{'value=1736758871659, type=class java.lang.Long, identifying=tru
e'}}]]
2025-01-13T10:01:11.964+01:00  INFO 216725 --- [exercice-2] [nio-8082-exec-1] o.s.b.c.l.s.TaskExecutorJobLauncher :
Job: [SimpleJob: [name=salesJob]] completed with the following parameters: [{'time':{'value=1736758871659, type=class j
ava.lang.Long, identifying=true'}}] and the following status: [COMPLETED] in 220ms
2025-01-13T10:01:17.159+01:00  DEBUG 216725 --- [exercice-2] [nio-8082-exec-2] org.hibernate.SQL :
select ssl_0.id,ssl_0.average_revenue_per_product,ssl_0.category,ssl_0.total_products,ssl_0.total_revenue from sales_su
mmmary ssl_0
Hibernate: select ssl_0.id,ssl_0.average_revenue_per_product,ssl_0.category,ssl_0.total_products,ssl_0.total_revenue fro
m sales_summary ssl_0
^[[^B
X  main*+  @ 0 0 0 0  Java: Ready
```

Exercice 3 : Kafka Streams

1. Développez une application Kafka Streams qui lit les données des véhicules depuis un topic Kafka nommé `vehicle_data`, filtre les véhicules qui dépassent la vitesse autorisée (par exemple, 80 km/h), puis génère un message d'alerte dans un topic Kafka nommé `speed_alerts`.

```
// VehicleData.java
@Data
@AllArgsConstructor
@NoArgsConstructor
public class VehicleData {
    private String vehicleId;
```

```

        private double speed;
        private double latitude;
        private double longitude;
        private double distanceToObstacle;
        private LocalDateTime timestamp;

        public static VehicleData fromString(String value) {
            String[] parts = value.split("\\|");
            return new VehicleData(
                parts[0],
                Double.parseDouble(parts[1]),
                Double.parseDouble(parts[2]),
                Double.parseDouble(parts[3]),
                Double.parseDouble(parts[4]),
                LocalDateTime.parse(parts[5], DateTimeFormatter.ofPattern("yyyy-MM-dd HH:mm:ss"))
            );
        }
    }
}

// KafkaConfig.java
@Configuration
@EnableKafka
@EnableKafkaStreams
public class KafkaConfig {
    @Bean
    public KafkaStreamsConfiguration kStreamsConfig() {
        Map<String, Object> props = new HashMap<>();
        props.put(APPLICATION_ID_CONFIG, "vehicle-monitoring-app");
        props.put(BOOTSTRAP_SERVERS_CONFIG, "localhost:9092");
        props.put(DEFAULT_KEY_SERDE_CLASS_CONFIG,
            Serdes.String().getClass().getName());
        props.put(DEFAULT_VALUE_SERDE_CLASS_CONFIG,
            Serdes.String().getClass().getName());
        return new KafkaStreamsConfiguration(props);
    }
}

// VehicleMonitoringService.java (partie 1)
@Service
public class VehicleMonitoringService {
    private static final double SPEED_LIMIT = 80.0;

    @Autowired
    public void processStream(StreamsBuilder streamsBuilder) {
        KStream<String, String> inputStream =
            streamsBuilder.stream("vehicle_data");

        // Traitement des alertes de vitesse
        inputStream
            .filter((key, value) -> {
                try {

```

```

        VehicleData data = VehicleData.fromString(value);
        return data.getSpeed() > SPEED_LIMIT;
    } catch (Exception e) {
        return false;
    }
})
.mapValues(value -> {
    VehicleData data = VehicleData.fromString(value);
    return String.format("%s|%.1f|%.6f|%.6f|0verspeeding|s",
        data.getVehicleId(), data.getSpeed(),
        data.getLatitude(), data.getLongitude(),
        data.getTimestamp());
})
.to("speed_alerts");
}
}

```

Configuration Kafka et création des topics :

```

# Création du réseau Docker
sudo docker network create kafkanet

# Démarrage de Zookeeper
sudo docker run -d --name zookeeper --network kafkanet \
    -e ZOOKEEPER_CLIENT_PORT=2181 \
    -e ZOOKEEPER_TICK_TIME=2000 \
    confluentinc/cp-zookeeper:7.3.0

# Démarrage de Kafka
sudo docker run -d --name kafka --network kafkanet -p 9092:9092 \
    -e KAFKA_BROKER_ID=1 \
    -e KAFKA_ZOOKEEPER_CONNECT=zookeeper:2181 \
    -e KAFKA_ADVERTISED_LISTENERS=PLAINTEXT://localhost:9092 \
    -e KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR=1 \
    -e KAFKA_TRANSACTION_STATE_LOG_MIN_ISR=1 \
    -e KAFKA_TRANSACTION_STATE_LOG_REPLICATION_FACTOR=1 \
    confluentinc/cp-kafka:7.3.0

# Création des topics
sudo docker exec -it kafka kafka-topics --create \
    --bootstrap-server localhost:9092 \
    --topic vehicle_data --partitions 1 --replication-factor 1

sudo docker exec -it kafka kafka-topics --create \
    --bootstrap-server localhost:9092 \
    --topic speed_alerts --partitions 1 --replication-factor 1

```

Test avec un véhicule en excès de vitesse :

```

# Envoi d'un message de test (vitesse = 135 km/h)
echo "vehicle789|135|48.8738|2.2950|20|2024-01-13 10:46:00" | \

```

```

sudo docker exec -i kafka kafka-console-producer \
--broker-list localhost:9092 --topic vehicle_data

# Vérification des alertes
sudo docker exec -it kafka kafka-console-consumer \
--bootstrap-server localhost:9092 \
--topic speed_alerts --from-beginning --max-messages 1

```

Résultat des alertes de vitesse :

```

# Sortie de la commande
1234|85.0|37.774900|-122.419400|Overspeeding|2024-01-13T10:15:00Z
Processed a total of 1 messages

```

2. Modifiez l'application Kafka Streams pour détecter les véhicules qui ont une distance au prochain obstacle inférieure à un certain seuil (par exemple, 5 mètres) et publier un message d'alerte dans un topic Kafka nommé `obstacle_alerts`.

```

// VehicleMonitoringService.java (partie 2)
@Service
public class VehicleMonitoringService {
    private static final double SPEED_LIMIT = 80.0;
    private static final double MIN_SAFE_DISTANCE = 5.0;

    @Autowired
    public void processStream(StreamsBuilder streamsBuilder) {
        KStream<String, String> inputStream =
streamsBuilder.stream("vehicle_data");

        // ... code précédent pour les alertes de vitesse ...

        // Traitement des alertes d'obstacles
        inputStream
            .filter((key, value) -> {
                try {
                    VehicleData data = VehicleData.fromString(value);
                    return data.getDistanceToObstacle() < MIN_SAFE_DISTANCE;
                } catch (Exception e) {
                    return false;
                }
            })
            .mapValues(value -> {
                VehicleData data = VehicleData.fromString(value);
                return String.format("%s|%.1f|Obstacle too close!",
                    data.getVehicleId(), data.getDistanceToObstacle());
            })
            .to("obstacle_alerts");
    }
}

```

Création du topic pour les alertes d'obstacles :

```
# Création du topic obstacle_alerts
sudo docker exec -it kafka kafka-topics --create \
  --bootstrap-server localhost:9092 \
  --topic obstacle_alerts --partitions 1 --replication-factor 1

# Test avec un véhicule trop proche d'un obstacle
echo "vehicle101|90|48.8584|2.2945|3|2024-01-13 10:47:00" | \
  sudo docker exec -i kafka kafka-console-producer \
  --broker-list localhost:9092 --topic vehicle_data

# Vérification des alertes d'obstacles
sudo docker exec -it kafka kafka-console-consumer \
  --bootstrap-server localhost:9092 \
  --topic obstacle_alerts --from-beginning --max-messages 1
```

Résultat des alertes d'obstacles :

```
# Sortie de la commande
5678|3.5|Obstacle too close!
Processed a total of 1 messages
```

Autres tests effectués :

```
# Test avec un véhicule dans les limites normales
echo "vehicle456|85|48.8584|2.2945|15|2024-01-13 10:45:00" | \
  sudo docker exec -i kafka kafka-console-producer \
  --broker-list localhost:9092 --topic vehicle_data

# Test avec un véhicule en excès de vitesse
echo "vehicle789|135|48.8738|2.2950|20|2024-01-13 10:46:00" | \
  sudo docker exec -i kafka kafka-console-producer \
  --broker-list localhost:9092 --topic vehicle_data

# Test avec un véhicule trop proche d'un obstacle
echo "vehicle101|90|48.8584|2.2945|3|2024-01-13 10:47:00" | \
  sudo docker exec -i kafka kafka-console-producer \
  --broker-list localhost:9092 --topic vehicle_data

# Test avec un véhicule en violation des deux conditions
echo "vehicle202|140|48.8584|2.2945|2|2024-01-13 10:48:00" | \
  sudo docker exec -i kafka kafka-console-producer \
  --broker-list localhost:9092 --topic vehicle_data
```

Vérification du fonctionnement de l'application :

```
# Liste des topics créés
sudo docker exec -it kafka kafka-topics --list --bootstrap-server
localhost:9092
```

Résultat :
obstacle_alerts
speed_alerts
vehicle_data

Images Kafka Streams :

```
$ sudo docker network create kafkanet
```

```
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$ sudo docker network create k
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$ sudo docker network create k
afkanet
[sudo] password for tr:
Sorry, try again.
[sudo] password for tr:
091d648261ea12b8630bf09fde5e951d7419422882b696d8f516d553f71e18f6
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$
```

```
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$ sudo docker run -d --name zookeeper --ne
twork kafkanet -e ZOOKEEPER_CLIENT_PORT=2181 -e ZOOKEEPER_TICK_TIME=2000 confluentinc/cp-zookeeper:7.3.0
Unable to find image 'confluentinc/cp-zookeeper:7.3.0' locally
7.3.0: Pulling from confluentinc/cp-zookeeper
d5d2e87c6892: Pull complete
008dba906bf6: Pull complete
bfeaabe01655: Pull complete
2cb7eb0f5666: Pull complete
f70f416c6ce7: Pull complete
bc67d000e59b: Pull complete
d6e744651f37: Pull complete
0427d86fae81: Pull complete
4108e73e61e1: Pull complete
ac5563423559: Pull complete
fa08a06f385f: Pull complete
bddb49e2fc4d: Pull complete
Digest: sha256:3ace7c3475a58afa128d0ee05bb9a06474b99b48c9d536c33144133637786f7f
```

```
$ sudo docker run -d --name kafka --network kafkanet -p 9092:9092 -e KAFKA_BROKER_ID=1 -e
KAFKA_ZOOKEEPER_CONNECT=zookeeper:2181 -e KAFKA_ADVERTISED_LISTENERS=PLAINTEXT://localhost:9092 -e
KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR=1 -e KAFKA_TRANSACTION_STATE_LOG_MIN_ISR=1 -e
KAFKA_TRANSACTION_STATE_LOG_REPLICATION_FACTOR=1 confluentinc/cp-kafka:7.3.0
```

```
008dba906bf6: Already exists
bfeaabe01655: Already exists
2cb7eb0f5666: Already exists
f70f416c6ce7: Already exists
bc67d000e59b: Already exists
d6e744651f37: Already exists
0427d86fae81: Already exists
4108e73e61e1: Already exists
ac5563423559: Already exists
d32323e291f3: Pull complete
ee69ff430d89: Pull complete
Digest: sha256:06e5d17d6c5163f3084ffb3c0a5f7f87ba87f35cc4739b598500b3db45dce35b
Status: Downloaded newer image for confluentinc/cp-kafka:7.3.0
docker: Error response from daemon: Conflict. The container name "/kafka" is already in use by container "5
28eeb2373d9da4f83c936f00646aa658cd43d9157c55c10bcd9dcffa81e9802". You have to remove (or rename) that conta
iner to be able to reuse that name.
See 'docker run --help'.
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/ ^C Cancel generation
```



```
$ sudo docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED
838e1df65537	confluentinc/cp-kafka:7.3.0	"/etc/confluent/dock_..."	5 seconds ago
Up 4 seconds	kafka	0.0.0.0:9092->9092/tcp, :::9092->9092/tcp	
4631b1a006af	confluentinc/cp-zookeeper:7.3.0	"/etc/confluent/dock_..."	26 seconds ago
Up 25 seconds	zookeeper	2181/tcp, 2888/tcp, 3888/tcp	
f2fe676f2f79	bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8	"/entrypoint.sh /run_..."	About an hour ago
Up About an hour (healthy)	datanode2	9864/tcp	
c3da872c8cd8	bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8	"/entrypoint.sh /run_..."	About an hour ago
Up About an hour (healthy)	datanode1	9864/tcp	

```
$ sudo docker exec -it kafka kafka-topics --create --topic vehicle_data --bootstrap-server localhost:9092 --partitions 1
--replication-factor 1 && sudo docker exec -it kafka kafka-topics --create --topic speed_alerts --bootstrap-server
localhost:9092 --partitions 1 --replication-factor 1 && sudo docker exec -it kafka kafka-topics --create --topic
obstacle_alerts --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1

do docker exec -it kafka kafka-topics --create --topic speed_alerts --bootstrap-server localhost:9092 --par
titions 1 --replication-factor 1 && sudo docker exec -it kafka kafka-topics --create --topic vehicle_data --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1 && su
do docker exec -it kafka kafka-topics --create --topic speed_alerts --bootstrap-server localhost:9092 --par
titions 1 --replication-factor 1 && sudo docker exec -it kafka kafka-topics --create --topic obstacle_alert
s --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide.
To avoid issues it is best to use either, but not both.
Created topic vehicle_data.
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide.
To avoid issues it is best to use either, but not both.
Created topic speed_alerts.
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide.
To avoid issues it is best to use either, but not both.
Created topic obstacle_alerts.
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$
```

```
$ sudo docker exec -it kafka kafka-topics --list --bootstrap-server localhost:9092

tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$ sudo docker exec -it kafka kafka-topics
--list --bootstrap-server localhost:9092
obstacle_alerts
speed_alerts
vehicle_data
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam$
```

```
$ sleep 10 && echo "vehicle123|120|48.8566|2.3522|5|2024-01-13 10:40:00" | sudo docker exec -i kafka kafka-console-producer
--broker-list localhost:9092 --topic vehicle_data

tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam/exercise_3$ sleep 10 && echo "vehicle123|120|48.8566|2.3522|5|
2024-01-13 10:40:00" | sudo docker exec -i kafka kafka-console-producer --broker-list localhost:9092 --topic vehicle_data
[sudo] password for tr:
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam/exercise_3$
```

```
$ echo "vehicle101|90|48.8584|2.2945|3|2024-01-13 10:47:00" | sudo docker exec -i kafka kafka-console-producer --broker-list
localhost:9092 --topic vehicle_data
```

```
echo "vehicle202|140|48.8584|2.2945|2|2024-01-13 10:48:00" | sudo docker exec -i kafka kafka-console-producer --broker-list
localhost:9092 --topic vehicle_data
```

```
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam/exercice-3$ sudo docker exec -it kafka kafka-console-consumer
--bootstrap-server localhost:9092 --topic speed_alerts --from-beginning --max-sudo docker exec -it kafka kafka-console-consumer
--bootstrap-server localhost:9092 --topic speed_alerts --from-beginning --max-messages 2
1234|85.0|37.774900|-122.419400|Overspeeding|2024-01-13T10:15:00Z
```

```
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam/exercice-3$ sudo docker exec -it kafka kafka-console-consumer
--bootstrap-server localhost:9092 --topic speed_alerts --from-beginning --max-sudo docker exec -it kafka kafka-console-consumer
--bootstrap-server localhost:9092 --topic speed_alerts --from-beginning --max-messages 2
1234|85.0|37.774900|-122.419400|Overspeeding|2024-01-13T10:15:00Z
^C
Processed a total of 1 messages
tr@another-linux-user:~/Documents/myStudies/SDIA/S3/big_data/exam/exercice-3$ sudo docker exec -it kafka kafka-console-consumer
--bootstrap-server localhost:9092 --topic obstacle_alerts --from-beginning --max-sudo docker exec -it kafka kafka-console-consumer
--bootstrap-server localhost:9092 --topic obstacle_alerts --from-beginning --max-messages 2
5678|3.5|Obstacle too close!
```

```
2025-01-13T10:40:26.573+01:00 INFO 239343 --- [exercice-3] [-StreamTh
read-1] o.a.k.c.c.internals.SubscriptionState : [Consumer clientId=
vehicle-monitoring-app-44f13180-0c10-4b02-83a9-20e1025d11bf-StreamThre
ad-1-consumer, groupId=vehicle-monitoring-app] Resetting offset for pa
rtition vehicle_data-0 to position FetchPosition{offset=0, offsetEpoch
=Optional.empty, currentLeader=LeaderAndEpoch{leader=Optional[localhos
t:9092 (id: 1 rack: null)], epoch=0}}.
2025-01-13T10:40:26.650+01:00 INFO 239343 --- [exercice-3] [-StreamTh
read-1] o.a.k.c.c.internals.ConsumerCoordinator : [Consumer clientId=
vehicle-monitoring-app-44f13180-0c10-4b02-83a9-20e1025d11bf-StreamThre
ad-1-consumer, groupId=vehicle-monitoring-app] Found no committed offs
et for partition vehicle_data-0
2025-01-13T10:40:26.656+01:00 INFO 239343 --- [exercice-3] [-StreamTh
read-1] o.a.k.s.processor.internals.StreamTask : stream-thread [vehi
cle-monitoring-app-44f13180-0c10-4b02-83a9-20e1025d11bf-StreamThread-1
] task [0_0] Restored and ready to run
2025-01-13T10:40:26.657+01:00 INFO 239343 --- [exercice-3] [-StreamTh
read-1] o.a.k.s.p.internals.StreamThread : stream-thread [vehi
cle-monitoring-app-44f13180-0c10-4b02-83a9-20e1025d11bf-StreamThread-1
] State transition from PARTITIONS ASSIGNED to RUNNING
2025-01-13T10:40:26.660+01:00 INFO 239343 --- [exercice-3] [-StreamTh
read-1] org.apache.kafka.streams.KafkaStreams : stream-client [vehi
cle-monitoring-app-44f13180-0c10-4b02-83a9-20e1025d11bf] State transit
ion from REBALANCING to RUNNING
2025-01-13T10:42:23.287+01:00 INFO 239343 --- [exercice-3] [-StreamTh
read-1] o.a.k.s.p.internals.StreamThread : stream-thread [vehi
cle-monitoring-app-44f13180-0c10-4b02-83a9-20e1025d11bf-StreamThread-1
] Processed 7 total records, ran 0 punctuators, and committed 3 total
tasks since the last update
2025-01-13T10:44:23.288+01:00 INFO 239343 --- [exercice-3] [-StreamTh
```